

# ARI 410/510 Problem Set Guidelines

University of Michigan-Flint – Fall 2026

(See course Canvas page for due date)

## What to do

There are several stages we will go through for these problems:

**Solo:** Complete the problems on your own and *submit your answers before the start of class on the due date*. Graded on completion and effort. **Answer the questions on your own (do not use AI tools to do them for you)**, for now, based on your understanding of the course material.

**Group:** At the start of class on the due date, in small groups, come to a consensus about the answers. Remember who you worked with. Graded based on engagement with group members.

**Class:** For each question, one student will be randomly selected to present and justify the solution. The entire class may ask questions and work together as necessary to reach a solution. Graded based on participation, effort, **and demonstrated understanding of the core premises of the problem**.

**Final submission & Bonus:** After class, make a second, updated submission with annotations regarding what changed and why. List the names of the group members you worked with and certify their engagement in the discussion. Some problem sets will have a bonus question that can be completed at this stage. Submit by the “final submission” due date. Graded based on correctness of final answers and thoughtful explanations of changes made.

## What to submit to Canvas

**Solo submission [first deadline]:** A document with your solutions. It can be a copy of this document, or one that only contains your solutions. A PDF editor, word processor, Google Docs, or a photo of neat handwritten work are all fine, as long as your answers are clearly readable.

**Final submission [second deadline]:** A copy of your Solo solutions with any updates in a visually distinct way (e.g., a different color, or a comment feature), where updates include **the change and why you made it**. Also attach the completed group-member evaluation (plus answers to any bonus question(s)).

## Resources

You may use resources from class or online, but the answers you submit must represent your own work and effort. **Do not use generative AI tools to directly answer the questions for you.** It is fine to ask an AI for explanations of concepts (e.g., “what is the difference between a weight and a bias term?”) or for reference information. But for a problem that asks you to derive or compute something, having an AI produce that derivation or computation counts as directly answering it. In the end, you need to be able to solve the problem on your own.

## Other notes

**Questions / Clarifications:** ask in class or on Discord (#homework-help). **Demonstrating effort:** showing the steps you followed may be required to receive credit. **Assumptions:** if you need to make a reasonable assumption about something unclear, state it in your answer. **Levels:** most problems are shared. Where a problem forks, a [410] and a [510] marker show each track; do the one for your section. (410 students are welcome to attempt the 510 version and include it.)

## Problem Set 2: Trail-Ready Motors

Trail-Ready Motors is a small shop in Michigan that buys used Jeeps, Subarus, and pickups, rebuilds some of them for overlanding, and resells them. The owners are starting several machine learning projects to help run the business.

### Problem 1. Where should the labels come from?

Every one of the shop's projects needs labeled data, and there are four main ways you might think about getting labels, like we talked about in class:

- **Historical data:** records of what actually happened, such as the shop's repair records for vehicles it sold in past years.
- **Human annotation:** people (e.g., shop's mechanics or staff) look at each example and label it.
- **Generated labels:** an AI model, such as an LLM, reads each example and outputs a label. These are then used to train another model.
- **Weak supervision:** simple rules or heuristics are combined to produce labels automatically.

Below are five of the shop's tasks. For each one, choose the label source you think fits best (you can combine sources if you explain how), say why in a sentence or two, and name the biggest risk of that choice for this task.

| Task                                                                                                                              | Label source | Why this source | Biggest risk |
|-----------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|--------------|
| <b>Major repair:</b> will this vehicle need a repair over \$1,500 within a year of being sold?                                    |              |                 |              |
| <b>Rust:</b> does an inspection photo show rust damage?                                                                           |              |                 |              |
| <b>Misleading listings:</b> is a seller's listing description misleading about the vehicle's condition?                           |              |                 |              |
| <b>Duplicates:</b> is a listing on a used-car marketplace a duplicate or spam post?                                               |              |                 |              |
| <b>Search tags:</b> which features (lift kit, roof rack, winch, and so on) does a listing mention, for the shop's website search? |              |                 |              |

**(b)** Suppose the shop trains its rust detector on ChatGPT-generated labels because they're cheap and effective. Why should the test set (at a minimum) still be labeled by mechanics or other experts?

## Problem 2. Do the labelers agree?

For the misleading-listings task, the shop decided to have two staff members, Rosa and Dev, independently read the same 20 listing descriptions and label each one “misleading: yes or no.” Their labels compare like this:

|           | Dev: Yes | Dev: No |
|-----------|----------|---------|
| Rosa: Yes | 6        | 3       |
| Rosa: No  | 1        | 10      |

**(a)** Compute the observed agreement,  $p_o$  (the fraction of vehicles they labeled the same way).

**(b)** Compute the agreement expected by chance,  $p_e$ . (Use how often each person says “yes” and “no” overall.)

**(c)** Compute Cohen’s kappa:

$$\kappa = (p_o - p_e) / (1 - p_e)$$

**(d)** How would you describe this level of agreement? Would you be comfortable using these labels to train a model as they are? Suggest one concrete step the shop could take to improve agreement (not just “label more data”).

### Problem 3. Grouping the inventory

To plan its upcoming marketing campaigns, the shop wants to group its vehicles without using any labels. The shop rates six vehicles on two scores from 0 to 10:

| Vehicle | Off-road capability | Daily comfort |
|---------|---------------------|---------------|
| A       | 3                   | 6             |
| B       | 2                   | 9             |
| C       | 4                   | 9             |
| D       | 8                   | 1             |
| E       | 7                   | 1             |
| F       | 6                   | 4             |

Let's assume we are running K-means with  $K = 2$ , starting with the centroids at vehicle A ( $c_1 = (3, 6)$ ) and vehicle B ( $c_2 = (2, 9)$ ). **Use squared Euclidean distance, so you don't need any square roots.**

**(a)** Below is each vehicle's squared distance to both centroids. In the last column, assign each vehicle to the closer cluster. Below the table, compute the two new centroids. Show your work.

| Vehicle | Distance to $c_1$ | Distance to $c_2$ | Cluster Assignment |
|---------|-------------------|-------------------|--------------------|
| A (3,6) | 0                 | 10                |                    |
| B (2,9) | 10                | 0                 |                    |
| C (4,9) | 10                | 4                 |                    |
| D (8,1) | 50                | 100               |                    |
| E (7,1) | 41                | 89                |                    |
| F (6,4) | 13                | 41                |                    |

**(b)** Given these new centroids, which vehicle(s) change(s) clusters, and why?

(c) Compute the final **inertia** (the sum of each vehicle's squared distance to its assigned centroid) using the centroids you computed in part (a).

(d) If you ran K-means with  $K = 6$  on these six vehicles, what would the inertia be? Why doesn't that make  $K = 6$  the best choice, and how would you choose  $K$  instead?

#### Problem 4. Which feature should the tree split on first?

The shop has labeled eight vehicles and wants to build a decision tree. Each vehicle has two features: whether the inspection found rust, and whether its mileage is high (over 150,000 miles).

| Vehicle | Rust | Mileage | Major repair? |
|---------|------|---------|---------------|
| V1      | Yes  | High    | Yes           |
| V2      | No   | Low     | No            |
| V3      | Yes  | High    | Yes           |
| V4      | No   | High    | No            |
| V5      | Yes  | Low     | No            |
| V6      | Yes  | High    | Yes           |
| V7      | No   | High    | Yes           |
| V8      | No   | Low     | No            |

**Values you may need:**

$\log_2(1) = 0$ ,  $\log_2(1/2) = -1$ ,  $\log_2(1/4) = -2$ ,  $\log_2(3/4) \approx -0.415$ ,  $\log_2(1/5) \approx -2.322$ ,  $\log_2(4/5) \approx -0.322$ .  
By convention, assume that  $0 \cdot \log_2(0) = 0$  even if  $\log_2(0)$  itself is undefined.

(a) Compute the entropy of the label,  $H(\text{Repair})$ .

**(b)** Compute the conditional entropies  $H(\text{Repair} \mid \text{Rust})$  and  $H(\text{Repair} \mid \text{Mileage})$ .

**(c)** Compute the information gain of each feature. Which feature should the tree split on first, and why?

**[410] (d)** Draw the depth-1 tree (one split) using the feature you chose. Label each leaf with the prediction it would make and how many training vehicles of each class reach it. What is the tree's accuracy on these eight vehicles?

**[510] (d)** The shop's spreadsheet also has a Listing ID column, a unique number for each vehicle. Compute the information gain of splitting on Listing ID. Would this be a good first feature to split on? Explain why or why not?